

Linear Algebra I: HW1 Partial Solution

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1. TFFFTFFFTT

2. No. From VS3, we deduce that $0_V = (0, 1)$. However, define $v = (a, 0)$, then for any $v' = (x, y)$,

$$v + v' = (a + x, 0 \cdot y) = (a + x, 0) \neq 0_V.$$

Hence v does not have any inverse element, which violates VS4. Therefore, V is not a vector space.

3. Check VS1 to VS8 one by one carefully.

4. TFTFT

5. Note that an upper triangular matrix (when $m = n$) is of the form

$$\begin{pmatrix} * & * & \cdots & * & * \\ 0 & * & \cdots & * & * \\ & & \ddots & * & * \\ 0 & 0 & \cdots & * & * \\ 0 & 0 & \cdots & 0 & * \end{pmatrix}$$

and

$$\begin{pmatrix} * & * & \cdots & * & * \\ 0 & * & \cdots & * & * \\ & & \ddots & * & * \\ 0 & 0 & \cdots & * & * \\ 0 & 0 & \cdots & 0 & * \end{pmatrix} + \begin{pmatrix} * & * & \cdots & * & * \\ 0 & * & \cdots & * & * \\ & & \ddots & * & * \\ 0 & 0 & \cdots & * & * \\ 0 & 0 & \cdots & 0 & * \end{pmatrix} = \begin{pmatrix} * & * & \cdots & * & * \\ 0 & * & \cdots & * & * \\ & & \ddots & * & * \\ 0 & 0 & \cdots & * & * \\ 0 & 0 & \cdots & 0 & * \end{pmatrix}$$

Similar argument can be also applied to the case $m \neq n$. [UPDATE: A more elegant way is to use the equality \$\(A + B\)_{ij} = A_{ij} + B_{ij}\$ for matrices \$A, B\$ of the same size. With this we can get rid of the tiring case by case argument. Thanks for pointing that out!](#)

6. (b) If W is a subspace of V and W contains W_1 and W_2 , then given $w_1 \in W_1 \subseteq W$, $w_2 \in W_2 \subseteq W$, we must have $w_1 + w_2 \in W$ because W is a subspace. Since w_1, w_2 are chosen arbitrarily, $W_1 + W_2 \subseteq W$.

7. For the case $m = n$, we have

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1,n-1} & a_{1,n} \\ a_{21} & a_{22} & \cdots & a_{2,n-1} & a_{2,n} \\ & & \ddots & & \\ a_{n-1,1} & a_{n-1,2} & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n-1} & a_{n,n} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1,n-1} & a_{1,n} \\ 0 & a_{22} & \cdots & a_{2,n-1} & a_{2,n} \\ & & \ddots & & \\ 0 & 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & 0 & \cdots & 0 & a_{n,n} \end{pmatrix} + \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ a_{21} & 0 & \cdots & 0 & 0 \\ & & \ddots & & \\ a_{n-1,1} & a_{n-1,2} & \cdots & 0 & 0 \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n-1} & 0 \end{pmatrix}$$

The first matrix is in W_1 , and the second one is in W_2 . Furthermore,

$$W_1 \cap W_2 = \{A \in M_n(F) \mid A_{ij} = 0 \text{ for all } i, j\} = \{0\}.$$

Similar argument can be applied to the case $m \neq n$.

UPDATE: Given $M \in M_{m \times n}(F)$, we define $M_1, M_2 \in M_{m \times n}(F)$ by

$$(M_1)_{ij} = \begin{cases} 0 & \text{if } i > j \\ M_{ij} & \text{if } i \leq j \end{cases} \quad (M_2)_{ij} = \begin{cases} 0 & \text{if } i \leq j \\ M_{ij} & \text{if } i > j \end{cases}$$

Then $M_1 \in W_1, M_2 \in W_2$ and $M = M_1 + M_2$. This argument can apply to matrices of arbitrary size.